

SIMATS ENGINEERING

ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

Annexure A

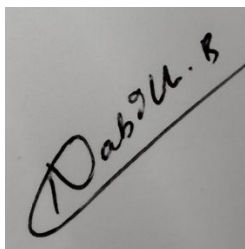
APPLICATION- BASED QUESTIONS

Code of Conduct:

I Nabilah Thahseen.B 1925260057(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in black ink on a light-colored surface. The signature is written in a cursive style and appears to read 'Nabilah Thahseen.B'.

APPLICATION- BASED QUESTIONS

Scenario: **Intercepted Suspicious Message** A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Deduce the most likely original message.

Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Understanding of the Concept

The given ciphertext “**GSRH RH Z HVXIVG**” has the characteristics of a **simple substitution cipher**. A suitable cipher for this message is the **Atbash cipher**.

The Atbash cipher is a substitution cipher in which the alphabet is reversed. Each letter is replaced by the letter at the corresponding position from the opposite end of the alphabet.

The substitution follows:

A	$\leftrightarrow$	Z
B	$\leftrightarrow$	Y
C	$\leftrightarrow$	X
D	$\leftrightarrow$	W
E	$\leftrightarrow$	V
F	$\leftrightarrow$	U
G	$\leftrightarrow$	T
H	$\leftrightarrow$	S
I	$\leftrightarrow$	R
J	$\leftrightarrow$	Q
K	$\leftrightarrow$	P
L	$\leftrightarrow$	O
M $\leftrightarrow$ N		

The same rule can be used for decryption because the Atbash transformation is reversible.

2. Identification of the Cipher

The ciphertext contains four words:

GSRH | RH | Z | HVXIVG

The word lengths are:

4 | 2 | 1 | 6

The same word structure is preserved after substitution. This is an important clue because a substitution cipher changes letters but does not change the number of letters in each word.

The one-letter word **Z** can reasonably decrypt to **A** using the Atbash rule.

Applying the reversed alphabet mapping to the other words produces meaningful English words.

Therefore, the most likely cipher is the **Atbash substitution cipher**

3. Step-by-Step Decryption

Atbash Rule

A ↔ Z, B ↔ Y, C ↔ X, D ↔ W, E ↔ V, F ↔ U, G ↔ T, H ↔ S, I ↔ R, J ↔ Q, K ↔ P, L ↔ O, M ↔ N

Now decrypt each character.

Ciphertext	Decrypted Letter
G	T
S	H
R	I
H	S
R	I
H	S
Z	A
H	S
V	E
X	C
I	R
V	E
G	T

Therefore:

GSRH → THIS

RH → IS

Z → A

HVXIVG → SECRET

Hence, the complete decrypted message is:

“THIS IS A SECRET”

4. Analysis of the Result

The decrypted message is **“THIS IS A SECRET.”**

The result can be validated using several observations:

1. The ciphertext has the word pattern **4–2–1–6**, and the decrypted message has the same word-length pattern.
2. The first word **GSRH** becomes **THIS**, which is a valid English word.
3. The second word **RH** becomes **IS**, which is grammatically meaningful.
4. The single-letter word **Z** becomes **A**, which is a valid English article.
5. The final word **HVXIVG** becomes **SECRET**, producing a meaningful six-letter word.
6. Every letter is consistently converted using the same Atbash substitution rule.
7. The complete decrypted sentence is meaningful and grammatically correct, which strongly validates the result.

5. Final Answer

Ciphertext: GSRH RH Z HVXIVG

Cipher Used: Atbash Substitution Cipher

Decrypted Message:

THIS IS A SECRET

Therefore, the most probable original message is **“THIS IS A SECRET.”**

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Understanding of the Concept

The keyword-based encryption technique used in this scenario is the **Vigenère cipher**.

The Vigenère cipher is a classical **polyalphabetic substitution cipher**. It uses a keyword to determine the shift applied to each character of the plaintext.

Unlike a simple substitution cipher, where the same plaintext letter always maps to the same ciphertext letter, the Vigenère cipher can produce different ciphertext letters for the same plaintext letter depending on the corresponding keyword character.

In this problem:

Plaintext = HELLO

Keyword = KEY

Since the plaintext contains five characters and the keyword contains only three characters, the keyword must be repeated.

Therefore:

HELLO

KEYKE

2. Repeating the Keyword

The keyword is:

KEY

To match the five characters of **HELLO**, it is repeated as:

KEYKE

The alignment is:

Position 1 2 3 4 5

Plaintext H E L L O

Keyword K E Y K E

Each plaintext character is encrypted using the corresponding keyword character.

3. Encryption Formula

Using the standard alphabet numbering:

A = 0, B = 1, C = 2, ..., Z = 25

The Vigenère encryption formula is:

$$C = (P + K) \bmod 26$$

Where:

- **P** = numerical value of the plaintext letter
- **K** = numerical value of the keyword letter
- **C** = numerical value of the ciphertext letter
- **mod 26** keeps the result within the range 0–25.

4. Step-by-Step Encryption

Character 1

Plaintext letter:

$$H = 7$$

Keyword letter:

$$K = 10$$

Calculation:

$$C = (7 + 10) \bmod 26$$

$$C = 17$$

17 corresponds to:

R

Therefore:

$$H \rightarrow R$$

Character 2

Plaintext letter:

$$E = 4$$

Keyword letter:

$$E = 4$$

Calculation:

$$C = (4 + 4) \bmod 26$$

$$C = 8$$

8 corresponds to:

I

Therefore:

$$E \rightarrow I$$

Character 3

Plaintext letter:

$$L = 11$$

Keyword letter:

$$Y = 24$$

Calculation:

$$C = (11 + 24) \bmod 26$$

$$C = 35 \bmod 26$$

$$C = 9$$

9 corresponds to:

J

Therefore:

L → J

Character 4

Plaintext letter:

$$L = 11$$

Keyword letter:

$$K = 10$$

Calculation:

$$C = (11 + 10) \bmod 26$$

$$C = 21$$

21 corresponds to:

V

Therefore:

L → V

Character 5

Plaintext letter:

$$O = 14$$

Keyword letter:

$$E = 4$$

Calculation:

$$C = (14 + 4) \bmod 26$$

$$C = 18$$

18 corresponds to:

S

Therefore:

O → S

5. Complete Encryption Table

Position	Plaintext P	P Value	Keyword K	K Value	Calculation	Ciphertext
1	H	7	K	10	$(7 + 10) \bmod 26 = 17$	R
2	E	4	E	4	$(4 + 4) \bmod 26 = 8$	I
3	L	11	Y	24	$(11 + 24) \bmod 26 = 9$	J

4	L	11	K	10	$(11 + 10) \bmod 26 = 21$ V
5	O	14	E	4	$(14 + 4) \bmod 26 = 18$ S

Therefore:

Plaintext: HELLO

Repeated Keyword: KEYKE

Ciphertext: RIJVS

6. Effect of the Repeating Keyword

The keyword **KEY** is repeated to match the length of the plaintext.

Therefore:

HELLO

is aligned with:

KEYKE

Each keyword character controls the shift applied to the corresponding plaintext character.

The first character **H** uses **K** and becomes **R**.

The second character **E** uses **E** and becomes **I**.

The third character **L** uses **Y** and becomes **J**.

The fourth character **L** uses **K** and becomes **V**.

The fifth character **O** uses **E** and becomes **S**.

An important observation is that the two **L** characters do not produce the same ciphertext character.

The first **L** uses **Y** and becomes **J**, while the second **L** uses **K** and becomes **V**.

This demonstrates how the repeating keyword changes the substitution applied to each character.

7. Analysis and Validation of the Result

The encryption result can be validated by checking every character independently.

The calculated ciphertext values are:

17, 8, 9, 21, 18

These values correspond to:

R, I, J, V, S

Therefore:

HELLO + KEYKE → RIJVS

All five characters have been encrypted using the same Vigenère formula, and the keyword has been correctly repeated to match the message length.

8. Final Answer

Plaintext: HELLO

Keyword: KEY

Repeated Keyword: KEYKE

Encryption Technique: Vigenère Cipher

Ciphertext:

RIJVS

Therefore, the encrypted form of “**HELLO**” using the keyword “**KEY**” is “**RIJVS.**”

CONCLUSION

Question 2 demonstrates the use of a **simple substitution cipher**, where the intercepted message “**GSRH RH Z HVXIVG**” is successfully decrypted using the Atbash technique to obtain “**THIS IS A SECRET.**”

Question 3 demonstrates a **keyword-based Vigenère encryption**, where “**HELLO**” is combined with the repeated keyword “**KEYKE**” to produce the ciphertext “**RIJVS.**”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation